

Task	Responsible	Start Date	End Date	Duration (in hours/points)	Status	Comments
S01: Sprint 1 (Weeks 1-2)						
Attend Week 1 support session	Brooke, Liam, Liza, Upite	24-Feb	24-Feb	1	Complete	Liam, Liza and Upite were unable to attend due to work commitments but have gone through and read the supporting pdfs
Email Sam to set up industry partner meeting	Liza	27-Feb	27-Feb	0.5	Complete	Organized initial and fortnightly meetings
Email Dimitri to set up catch up meeting	Liza	27-Feb	27-Feb	0.5	Complete	Organized zoom meeting
Attend industry partner meeting	Brooke, Liam, Liza, Upite	28-Feb	28-Feb	1	Complete	Upite was unavailable to attend due to work commitments
Attend meeting with Dimitri	Brooke, Liam, Liza, Upite	28-Feb	28-Feb	1	Complete	Liam and Upite unavailable to attend due to work and university commitments
Create Phase 2 sprint document	Liza	28-Feb	28-Feb	1	Complete	Created and uploaded document in shared folder
Spend time organizing shared folders	Liza	28-Feb	28-Feb	0.5	Complete	Worked for ~30 mins on setting up shared folder environment to keep it organized
Typing up notes from Sam and Dimitri meetings	Brooke, Liza	28-Feb	28-Feb	0.5	Complete	Uploaded notes to shared folder for all team members to review
Spend time planning and writing tasks into the sprint document	Liza	28-Feb	28-Feb	0.5	Complete	Spent ~30 mins organizing, planning, writing and delegating tasks in the sprint document
Organize team meeting	Liza	28-Feb	28-Feb	0.5	Complete	Quick meeting to discuss plans moving forward
Attend team meeting	Brooke, Liam, Liza, Upite	1-Mar	1-Mar	1	Complete	Brooke and Upite were unable to attend due to personal and work reasons
Update the risk registry	Liam, Liza	1-Mar	1-Mar	0.5	Complete	Liam: Reviewed previous risk register items, updating where needed. Liza: added a few more risks based on the nature of the project.
Upload notes from Sam/Dimitri and Liza/Liam meeting	Liza	1-Mar	1-Mar	0.5	Complete	Created meeting notes shared folder and uploaded in there
Fill out student details on project owner feedback form	Liza	1-Mar	1-Mar	0.5	Complete	Noted the team name, meeting time and place and attendees

Not Started
In Progress
Dependency Incomplete
Reassigned
Incomplete
Complete



Email Sam the feedback form	Liza	1-Mar	1-Mar	0.5	Complete	Sent through feedback form, reiterating its purpose and what is required from our industry partner.
Send team update with delegated tasks	Liza	1-Mar	1-Mar	0.5	Complete	Sent through tasks required for 1st tutor meeting in messenger chat for convenience
Spend time working on the sprint document	Liza	1-Mar	1-Mar	1.5	Complete	Spent ~1.5 hours planning tasks, delegating and recording into the sprint document
Prepare and upload meeting minutes from all Week 1 meetings	Liza	1-Mar	2-Mar	0.5	Complete	All meeting minutes uploaded to shared folder for weeks 1-2
Create and upload product backlog document	Liza	1-Mar	2-Mar	1	Complete	Uploaded to shared document
Investigate the best project management technique to record planning and progress	Liza	1-Mar	2-Mar	1	Complete	Spent ~1 hour researching project management techniques and how to best organize project data and processes.
Add user stories, epics, prioritizations, effort estimates and sprint numbers to product backlog	Liza	2-Mar	2-Mar	2	Complete	Came up with and recorded 12 user stories, as well as most of the epics and prioritizations.
Spend time working on the sprint document	Liza	2-Mar	2-Mar	1	Complete	Spent ~1 hour working on the sprint document
Create burndown chart	Liza	2-Mar	2-Mar	1.5	Complete	Created and uploaded burndown chart for sprint 1 to shared folder
Create and upload the project progress template	Liza	2-Mar	2-Mar	0.5	Complete	Uploaded to shared folder
Create Slack Page	Brooke	3-Mar	3-Mar	0.5	Complete	A slack page was created for streamlined discussions about certain topics so important messages don't get missed
Create new Trello board	Brooke	3-Mar	3-Mar	0.5	Complete	A Trello board was created for easy viewing of project tasks and to ensure everyone is staying on top of their work
Request close up ruck footage from Jake	Liam	4-Mar	4-Mar	0.5	Complete	Email sent to Jake requesting close up and wide angle, shorter length footage of rucks
Read through and understand AS1 project guide and rubric	Brooke, Liam, Liza, Upite	1-Mar	4-Mar	2	Complete	Sign your name when complete: Liza, Upite, Liam, Brooke
Read through and understand AS3 report guide and rubric	Brooke, Liam, Liza, Upite	1-Mar	4-Mar	2	Complete	Sign your name when complete: Liza, Liam, Upite, Brooke

Watch and understand the professional and ethical conduct video (if you haven't already)	Brooke, Liam, Liza, Upite	1-Mar	4-Mar	2	Complete	Sign your name when complete: Liza, Liam, Upite, Brooke
Add acceptance criteria to the user stories in product backlog	Liza	4-Mar	4-Mar	1	Complete	Added acceptance criteria for all user stories in the product backlog
Setting up GitHub and GitHub branches	Liam, Liza	4-Mar	4-Mar	1.5	Complete	Setting up work environment with different GitHub branches for different work
Assign user story points to each user story on product backlog	Upite	4-Mar	4-Mar	1	Complete	Rough draft with Must Have's being 5pts, Should Have's 3pt, Could Have's 1pt. All subject to change. For each section, all individual stories will be reviewed to better account for time and complexity
Updated phase 2 communication plan within team agreement	Brooke	3-Mar	4-Mar	0.5	Complete	Communication section within Team Contract has been updated
Complete an updated phase 2 team agreement plan	Upite	4-Mar	4-Mar	1	Complete	Rest of Team Contract has been updated
Complete an updated phase 2 communication plan	Upite	4-Mar	4-Mar	1.5	Complete	Written. Located in Project Managment Documents folder on OneDrive
Document individual progress in progress report	Brooke, Liam, Liza, Upite	4-Mar	4-Mar	2	Complete	Sign your name when complete: Brooke, Upite, Liza, Liam
Document team progress in progress report	Liam	4-Mar	4-Mar	1	Complete	Completed in dot points - ready to discuss in further detail during meeting if needed.
Update player tracking code from phase 1 and commit changes to GitHub	Liza	4-Mar	4-Mar	1	Complete	Committed changes to GitHub - code to detect players highlighting in box outline
Update ball tracking code from phase 1 and commit changes to GitHub	Liam	4-Mar	4-Mar	1	Complete	Committed changes to GitHub - code to detect ball highlighting in box outline
Write code for segmentation tracking and selected main starting point	Liam, Liza	4-Mar	4-Mar	1.5	Complete	Written code to track objects in a frame simultaneously and chose tracking method as a universal starting point for consistency
Write code for detecting main colours in image	Liam, Liza	4-Mar	4-Mar	2	Complete	Written code to detect k number of colours from a static image
Write code to detect main colour (jersey) within bounding box	Liza	4-Mar	4-Mar	3	Complete	Written code to detect the main colour within a bounding box (usually the colour of jersey) and change the colour of the bounding box to main detected colour
Manually train a small sample of ruck data by labelling images	Liam	4-Mar	4-Mar	3	Complete	Using Roboflow, manually label location/formation of rucks and surrounding players to create a labelled database to train model.
Complete velocity charts	Upite	4-Mar	4-Mar	1	Complete	See other page 'Team Velocity'. Contact me for any questions

Attend meeting with Wayne	Brooke, Liam, Liza, Upite	4-Mar	4-Mar	0.5	Complete	All members present
Take notes and upload meeting minutes (from Wayne meeting)	Brooke	4-Mar	4-Mar	0.5	Complete	Uploaded to shared folder
Attend team meeting	Brooke, Liam, Liza, Upite	4-Mar	4-Mar	1	Complete	Discussed agenda for the next 2 sprints
Ensure all team members have their own GitHub branch to work from	Liam	4-Mar	4-Mar	0.5	Complete	Signed when complete: Liam, Liza, Brooke, Upite
Go through each video from reds and screenshot images of where the lineup's occurred	Brooke	7-Mar	7-Mar	3	Complete	Each video was watched and recorded where the lineouts occurred. Then Screenshots were taken of each lineup
Find the outermost point of the bounding box	Liza	9-Mar	9-Mar	3	Complete	Able to detect furthest point based on the rectangular bounding box. Need to tune bounding box for more flexible parameters. Realized that need to determine the direction a player is facing for outermost point detection to make sense in ruck context
Manipulate video to pause and check location of players	Liza	9-Mar	9-Mar	0.5	Complete	Created ability to pause and resume video as they are playing to get a better look at the position of players. Might need to create functionality to manipulate this image into 2D to better visualize positions.
Update risk register to include natural disasters, such as the cyclone	Liza	9-Mar	9-Mar	0.5	Complete	Added new risk due to the current circumstances
Work on sprint document	Liza	9-Mar	9-Mar	1.5	Complete	Updated Week 2 sprint plan, merged the burndown chart into the same document for convenience
Create burndown chart for Sprint 1	Liza	9-Mar	9-Mar	0.5	Complete	Created based off total story points for this sprint (weeks 1 and 2)
Tune parameters of the bounding box to better reflect player movements	Liza	9-Mar	~	2	In Progress	Started to tweak parameters - at the moment, only able to adjust box to be slightly smaller but still not contoured. <i>Will move to next sprint as in progress and keep working.</i>
Obtain additional close-up and wide-angle ruck footage	Liam	4-Mar	~	0.5	In Progress	Request has been put forward to client, still awaiting response. <i>Will move into next sprint with the status in progress until a response from client is received.</i>
Take notes and upload meeting minutes (from team meeting)	Brooke	~	~	0.5	Incomplete	Due to cyclone and power outage, Brooke had no internet access for most of the week. <i>This task was moved into the next sprint.</i>
TOTAL POINTS	60.5 / 63.5 COMPLETED					
S02: Sprint 2 (Weeks 3 - 4)						
(from sprint 1) Tune parameters of the bounding box to better reflect player movements	Liza	10-Mar	10-Mar	3	Complete	After hours of research and trial and error, it appears that the best way to tune parameters is to annotate the dataset in Roboflow so it's contoured rather than rectangular. After the model is trained on this data, it should detect it more contoured.

Upload each image of lineout to Roboflow and begin to manually outline where lineout players are	Brooke	12-Mar	14-Mar	3	Complete	Recorded each stage of where a lineout happened on the videos provided by the reds, began to screenshot and then upload into Roboflow
Send Zoom link to Sam	Liam	14-Mar	14-Mar	0.5	Complete	Link sent.
Email Sam the feedback form	Liam	14-Mar	14-Mar	0.5	Complete	Received filled out form.
Attend meeting with Sam	Brooke, Liam, Liza, Upite	14-Mar	14-Mar	0.5	Complete	Upite unable to attend due to work commitments.
Work in Roboflow, Editing the images for the lineouts and get the model from the dataset	Brooke	14-Mar	16-Mar	3	Complete	Went through the images uploaded to Roboflow and edited the parameters of where the lineouts are and the positions of where the players are
Get masks of detected objects for higher accuracy (from static image)	Liza	16-Mar	16-Mar	1.5	Complete	Get masks of individual people detected in static image and combine them to display in a single window. May need to adapt to video.
Draw tighter bounding box around detected objects (from static image)	Liza	16-Mar	16-Mar	0.5	Complete	Draw tighter, more curved lines around detected objects for potentially higher accuracy. Although, may need to adapt as some body parts (hands, feet) may be missing.
Investigate translating 3D footage with key dots to 2D footage	Liza	16-Mar	16-Mar	0.5	Complete	After some research, realized this may not be necessary at all, as it makes more sense to implement with individual players rather than the whole ruck.
Colour code bounding box based on player team colours	Liza	16-Mar	16-Mar	2	Complete	Colour is detecting and colouring bounding box, however it is mostly returning black. Possibly because it is reading too many pixels from the background, however I'm not too sure why it's not detecting the dominant colour properly.
(from sprint 1) Take notes and upload meeting minutes (from team meeting on 4th March)	Brooke	16-Mar	16-Mar	0.5	Complete	Minutes for the meeting were recorded and uploaded to team shared folder
Take notes and upload meeting minutes (Sam meeting)	Brooke	16-Mar	16-Mar	0.5	Complete	Meetings minutes were completed and uploaded to team shared folder
Manually train a small sample of lineout data by labelling images	Upite	~	~	0	Dependency Incomplete	Dependency task: Complete in week 1 of sprint. Didn't know where to start, watched some videos on how to use tool then had trouble connecting to shared workspace. As a result, unable to see whether the model is accurate. Moved into next week of the same sprint.
Use additional footage to create larger dataset within Roboflow (lineouts)	Upite	~	~	0	Dependency Incomplete	Dependency task: Complete in week 1 of sprint. Didn't know where to start, watched some videos on how to use tool then had trouble connecting to shared workspace. As a result, unable to test accuracy and consistency of model. Moved into next week of the same sprint.
Train and implement custom model within object detection (lineouts)	Upite	~	~	0	Dependency Incomplete	Dependency task: Complete in week 1 of sprint. Didn't know where to start, watched some videos on how to use tool then had trouble connecting to shared workspace. As a result, unable to test outer point and colour detection with lineouts. Moved into next week of same sprint.
Use additional footage to create larger dataset within Roboflow (rucks)	Liam	~	~	0	Dependency Incomplete	Dependency task: Complete in week 1 of sprint. Didn't have time to complete. As a result, unable to test accuracy and consistency of model. Moved into next week of the same sprint.
Train and implement custom model within object detection (rucks) - polygon borders	Liam	~	~	0	Dependency Incomplete	Dependency task: Complete in week 1 of sprint. Didn't have time to complete. As a result, unable to test outer point and colour detection with rucks. Moved into next week of the same sprint.

Upload signed feedback form to shared folder	Liam	17-Mar	17-Mar	0.5	Complete	Uploaded.
Send follow up email to Jake requesting footage	Liam	18-Mar	18-Mar	0.5	Complete	Follow up email sent
Use polygon tool to train model on ruck detection instead of rectangle tool	Liam	18-Mar	18-Mar	0.5	Complete	Applied polygon border to initial ruck dataset
(from week 1 of sprint 2) Train and implement custom model within object detection (rucks) - with polygon borders	Liam	18-Mar	18-Mar	1	Complete	Trained custom YOLOv11 model with updated polygon bordered dataset
(from week 1 of sprint 2) Use additional footage to create larger dataset within Roboflow (rucks)	Liam	18-Mar	18-Mar	2.5	Complete	As a plan B whilst the client is providing more footage, downloaded pre-made Roboflow dataset with over 1600 images to train custom model (YOLOv11)
Test YOLOv11 model from Roboflow dataset	Liam	18-Mar	18-Mar	0.5	Complete	Test created model on a video
Adapt outermost rectangle point identification to outermost segmentation point detection	Liza	18-Mar	18-Mar	1	Complete	Code now can detect the outermost point not by measuring the furthestmost corner from centre, but use seg mask to detect outermost point in any position of the detectable object
Determine which way players are facing in a ruck	Liza	18-Mar	18-Mar	0.5	Complete	After some research, realized that this is not necessary if we combine the segmentation mask outermost point detection with colour detection to determine teams and orientation.
Document individual progress in progress report	Brooke, Liam, Liza, Upite	18-Mar	18-Mar	1	Complete	Liza and Upite completed individual progress report
Attend meeting with Wayne	Brooke, Liam, Liza, Upite	18-Mar	18-Mar	0.5	Complete	All members present.
Attend team meeting	Brooke, Liam, Liza, Upite	18-Mar	18-Mar	1	Complete	All members present. Set up in person meeting to catch up on dependency tasks and get all members on the same page.
Work on sprint document	Liza	19-Mar	19-Mar	0.5	Complete	Rearranged the sprint document to display time sensitive / dependent tasks in week 1 of the sprint, and all other tasks in week 2 so it's clear what needs to be done first.
Attend in person team study session	Brooke, Liam, Liza, Upite	21-Mar	21-Mar	3	Complete	Kelvin Grove Campus, 11am. Liza had to leave early due to work commitments.
(from week 1 of sprint 2) Manually train a small sample of lineout data by labelling images	Upite	21-Mar	21-Mar	1.5	Complete	Whilst waiting for dataset from IP, we are training the model on dataset that we sourced externally
(from week 1 of sprint 2) Use additional footage to create larger dataset within Roboflow (lineouts)	Upite	21-Mar	21-Mar	1	Complete	Whilst waiting for dataset from IP, we are training the model on dataset that we sourced externally

(from week 1 of sprint 2) Train and implement custom model within object detection (lineouts)	Brooke, Liam, Upite	21-Mar	21-Mar	3.5	Complete	Running it in Roboflow but Liam is helping to set up in VS code
Work on trying to fix the problem within python	Brooke, Upite, Liam	21-Mar	21-Mar	3	Complete	There was a problem within python where it wasn't training the model. We all tried to find different things to solve it. Liam managed to find a different way to train the model without going through python but we are still able to implement and use the model in python
Create custom YOLOv11 model for rucks	Liam	21-Mar	21-Mar	2	Complete	Initial model trained through VSCode using 5 epochs. None of our computers have a dedicated GPU so training took too long. Utilized Google Colabs instead to complete training on 20 epochs much more efficiently.
Create custom YOLOv11 model for lineouts	Liam	21-Mar	21-Mar	2	Complete	Followed process above to train custom YOLOv11 in Google Colabs
Tested custom models	Liam	21-Mar	21-Mar	0.5	Complete	Tested some preliminary images and videos using custom models
Merge all completed work into main branch	Liam	21-Mar	21-Mar	0.5	Complete	Merged.
Debugging colour detection code to correctly classify players into teams	Liza	18-Mar	23-Mar	4	Complete	Can now detect and sort teams based on colour. May need to adjust thresholds for brightness, distance and ROI regions to find optimal but it works very well on the few images I have tested.
Based on 2 main team colours, classify any other detectable person as 'other'	Liza	23-Mar	23-Mar	2	Complete	If the colour is far enough from the two main team colours (can be specified with a threshold value), the person is considered part of the 'other' team.
If person doesn't belong to team 1 or 2 based on colour, don't detect them at all.	Liza	23-Mar	23-Mar	0.5	Complete	Now these people are not detected with a bounding box at all. Could cause problems if, for example, a team and the ref are both wearing black. May need to edit this in the future if time permits.
Draw a horizontal line on the image to show the bounds of outermost point	Liza	23-Mar	23-Mar	0.5	Complete	After researching and experimenting with code, realized this doesn't necessarily work unless we are looking at the field from bird's eye view. After researching other ways to implement this, still not sure. Need to schedule meeting with Dimitri to discuss possibly steps.
Work on sprint document	Liza	23-Mar	23-Mar	1	Complete	Moved incomplete tasks to next week. Finalized next week's sprint. Assigned tasks.
Create burndown chart for Sprint 2	Liza	23-Mar	23-Mar	0.5	Complete	Created burndown chart
Research how to download model trained in Roboflow and run in Python	Upite	23-Mar	23-Mar	1	Complete	After troubleshooting in Visual Studio Code troubleshooting and multiple team members having issues, the solution was to use Google Colab
(from sprint 1) Obtain additional close-up and wide-angle ruck footage	Liam	4-Mar	~	0.5	In Progress	Awaiting client response. Plan B includes using publicly available footage and pre-trained models to test code with. This will be moved into next week.
Take notes and upload meeting minutes (team meeting)	Brooke	18-Mar	~	0.5	In Progress	Notes have been taken but not uploaded. This will not impact the progression of our project as notes are for reflection and can be uploaded later. This will be moved into the next sprint.
Research how to measure distance and draw lines	Upite	20-Mar	~	2	In Progress	Incomplete. This will be moved into the next sprint.

Take notes and upload meeting minutes (Wayne meeting)	Brooke	18-Mar	~	0.5	In Progress	Notes have been taken but not uploaded. This will not impact the progression of our project as notes are for reflection and can be uploaded later. <i>This will be moved into the next sprint.</i>
Measure and mark the time the ball leaves player's hands in lineout	Brooke, Upite	~	~	3	Incomplete	Both Upite and Brooke had trouble getting the Roboflow model to train within python. Help was sourced from Liam, however they didn't have time to complete this before the end of the week. <i>This will be moved into the next sprint</i>
TOTAL POINTS	52.5 / 59 COMPLETED					
S03: Sprint 3 (Week 5 - 6)						
Schedule meeting with Dimitri	Liam	24-Mar	24-Mar	0	Reassigned	Dependency task: complete in week 1. Liam was working so reassigned to Liza. Doesn't impact progression of project.
(from sprint 2) Take notes and upload meeting minutes (Wayne meeting)	Brooke	26-Mar	26-Mar	0.5	Complete	Complete in week 1 before it's forgotten. Waiting for Brooke to upload notes to shared folder. Uploaded.
(from sprint 2) Take notes and upload meeting minutes (team meeting)	Brooke	26-Mar	26-Mar	0.5	Complete	Complete in week 1 before it's forgotten. Waiting for Brooke to upload notes to shared folder. Uploaded.
Schedule meeting with Dimitri	Liza	26-Mar	26-Mar	0.5	Complete	Asked to schedule meeting for Friday morning to discuss our progress, how to best identify offside and help Brooke with her VS code.
(from sprint 2) Document individual progress in progress report	Brooke	26-Mar	26-Apr	1	Complete	Completed
(from sprint 2) Research how to measure distance and draw lines	Upite	27-Mar	27-Mar	2	Complete	Dependency task: complete in week 1 Revisited to start next task
(from sprint 1) Obtain additional close-up and wide-angle ruck footage	Liam	4-Mar	28-Mar	0.5	Complete	Complete in week 1 so client is aware earlier. Had in person meeting with Dimitri and Jake where we obtained additional footage needed. Let them know about possible delays as the footage wasn't provided earlier.
Attend meeting with Dimitri and Jake	Brooke, Liam, Liza, Upite	28-Mar	28-Mar	1	Complete	Liam and Liza attended in person, Brooke and Upite joined over zoom. Jake also joined the meeting and provided additional footage needed. Dimitri provided feedback and tips on project progression.
Take notes and upload meeting minutes (Dimitri + Jake meeting)	Liza	28-Mar	28-Mar	0.5	Complete	Typed and uploaded to shared folder
Email Sam general summary of what we have accomplished last sprint and plan to do for this sprint + feedback form	Liam	28-Mar	28-Mar	0.5	Complete	Email sent
Cut ruck videos and upload to shared folder	Liam	28-Mar	28-Mar	1.5	Complete	Cut ruck offsides into individual videos and shared with team
Find a pre-trained ball detection dataset on Roboflow. Add custom images to dataset so it can be trained using our own footages as well. Mark up each of those	Brooke	28-Mar	28-Mar	3	Complete	Found a small example on Roboflow of the ball detection model. Added custom images from game of reds to the dataset. Annotated each of these images of where the ball was when it could be seen in shot and then got Liam to train the model and add it to our project

Train a new model using pre-trained ball detection dataset from Roboflow	Liam	28-Mar	28-Mar	0.5	Complete	Trained and shared with team
Adjust bounding box of pre-made dataset to use polygons	Brooke, Liam, Liza, Upite	28-Mar	28-Mar	0	Complete	Reevaluated the need for this at this stage in the project. Re-prioritized into a could have to improve model accuracy, given time at the end of the project.
Merge any finalized code to main	Liam	28-Mar	28-Mar	1	Complete	Encountered some errors along the way but figured out how to merge in the end. Updated everyone's branches.
Cut lineout videos and upload to shared folder + GitHub	Upite	29-Mar	29-Mar	1.5	Complete	Dependency task: complete in week 1. Complete and uploaded
Adjust colour detection and test on different game footage from wide angle (on images). Used YOLO model	Liza	30-Mar	30-Mar	0.5	Complete	Dependency task: complete in week 1. Altered the ROI to be in the top half and smaller area to focus on the jersey. Tested on images. Mostly accurate, however I noticed the further away the camera angle, the less people YOLO detects.
Write code for colour detection from video and test on wide angle footage (videos). Used YOLO model	Liza	30-Mar	30-Mar	0.5	Complete	Dependency task: complete in week 1. Now have code for colour detection from video, however video runs very slowly, possibly due to processing frames being computationally expensive.
Work on sprint document	Liza	30-Mar	30-Mar	1	Complete	Updated tasks
Create artefact agreement template	Brooke	31-Mar	31-Mar	0.5	Complete	Must be completed before meeting on Tuesday. Created and uploaded to shared folder
Start to fill in sections of the draft	Brooke	31-Mar	31-Mar	1	Complete	Must be completed before meeting on Tuesday. Filled in dot points for each section
Read through and take notes from the testing support session	Liza	1-Apr	1-Apr	1	Complete	Took notes on types of testing needed and uploaded to shared folder
Create testing plan template	Liza	1-Apr	1-Apr	1	Complete	Must be completed before meeting on Tuesday. Using online resources, created testing plan template for Agile framework and uploaded to shared folder
Fill in first 4 tests into testing plan	Liza	1-Apr	1-Apr	0.5	Complete	Must be completed before meeting on Tuesday. Filled in the first 4 tests to be done
Add 5 more tests into the testing plan	Liza	1-Apr	1-Apr	0.5	Complete	Must be completed before meeting on Tuesday. Added to testing plan document.
Work on sprint document	Liza	1-Apr	1-Apr	1.5	Complete	Updated tasks
Team meeting to discuss artefact agreement	Liam, Liza, Upite	1-Apr	1-Apr	0.5	Complete	Clarified questions about product deliverables

Create Release Plan	Upite	1-Apr	1-Apr	1	Complete	Created and uploaded to sprint excel
Draw a line for lineouts based on coordinates of the hooker	Upite	30-Mar	1-Apr	2.5	Complete	OpenCV stopped working on VS Code but started writing code unsure if working. Managed to draw line from hooker but uses grid coordinates as opposed to detecting where hooker/lineout is. Drawing was tested, works and is available on GitHub. Need to talk to Liam about how he went with the pitch detection.
Add more detailed dot points to complete artefact agreement draft	Upite	1-Apr	1-Apr	2	Complete	Must be completed before meeting on Tuesday. Filled in Roles and Responsibilities. Added to paragraphs where there was a [...] point. Went through guides and added Deliverables and Handover sections. Added scope paragraph
Create test ruck model python file	Liam	1-Apr	1-Apr	0.5	Complete	Must be completed before meeting on Tuesday. Created file in tests folder and started adding checks
Allow for console input functionality for file uploads and model types	Liam	1-Apr	1-Apr	1	Complete	Enable user to input file path name and model type instead of hard coding this information.
Writing a test to see if ruck model loads correctly	Liam	1-Apr	1-Apr	1	Complete	Must be completed before meeting on Tuesday. Complete
Writing tests to see if lineout and ball models load correctly	Liam	1-Apr	1-Apr	0.5	Complete	Must be completed before meeting on Tuesday. Complete
Attend meeting with Wayne	Brooke, Liam, Liza, Upite	1-Apr	1-Apr	0.5	Complete	All members present.
Take notes and upload meeting minutes (Wayne meeting)	Brooke	1-Apr	1-Apr	0	Reassigned	Unable to take notes during the meeting. Reassigned to Liza
Attend team meeting	Brooke, Liam, Liza, Upite	1-Apr	1-Apr	0.5	Complete	Brief meeting as it was late and Liza and Liam still needed to drive home after having meeting at uni.
Take notes and upload meeting minutes (team meeting)	Brooke	1-Apr	1-Apr	0	Complete	No agenda or project related information was discussed, so no minutes were produced.
Refactoring code	Liam	1-Apr	1-Apr	1	Complete	Several files within our code space were using identical code. Refactored code into smaller functions and moved reused functions into separate file
Added Docstring comments to refactored functions	Liam	1-Apr	1-Apr	0.5	Complete	Added header/docstring comments to each of the created functions for easy implementation at a later stage
Update main.py with user input for file path, model type and some model parameters	Liam	1-Apr	1-Apr	1	Complete	Updated the main.py file to allow easy testing of all different models and inference.
Update model inference code to output a Python generator	Liam	2-Apr	2-Apr	1	Complete	When calculating inference, the YOLO model defaults to outputting a list of all results, taking up lots of memory. Adjusted model parameters and refactored code accordingly to produce output into a generator instead. When running longest inference video, RAM usage dropped from approx. 6Gb to 40Mb
Get the ball model to work with the lineout model	Brooke	3-Apr	3-Apr	2	Complete	With the current models, they were combined to work together within the same python file.

Combine ball/lineout model with player detection code	Brooke	4-Apr	4-Apr	1	Complete	Using the two combined models of the ball and lineout, these were then combined with the player model so that we can use it to detect when the player releases the ball within the lineout
Split up the Reds video of a closer view	Brooke	4-Apr	4-Apr	2	Complete	After combining the three models together, when testing against the wide view footage, the models had a hard time detecting where the ball was still. It was decided to attempt to use footage from a closer view to see if that worked better.
Convert split up videos into GIF's to be able to use in python	Brooke	4-Apr	4-Apr	2	Complete	After splitting up the videos, each one had to be converted into a GIF. This took a while because there were quite a few and it took a little while for the software to convert them over
Take notes and upload meeting minutes (Wayne meeting)	Liza	5-Apr	5-Apr	0.5	Complete	Added to shared folder
Test colour detection with the new model for ruck detection	Liza	5-Apr	5-Apr	0.5	Complete	Tested colour detection with the ruck model and realized that it now detects the dominant colour of the players in the ruck, instead of the players outside the ruck. Need to edit code to detect 2 different objects - ruck and person.
Edit code to combine 2 models in one file - model to detect rucks and model to detect people	Liza	5-Apr	5-Apr	0.5	Complete	Adjusted code to include 2 models and 2 sets of results. Can now detect both the ruck and individual players in one video. Only issue is that the screen is busy. Perhaps an issue to solve in the next sprint.
Test on the ruck model with outermost point detection	Liza	5-Apr	5-Apr	0.5	Complete	Outermost point detection works with rucks detected. Shows the left and right outermost points.
Combine colour and outermost point detection codes	Liza	5-Apr	5-Apr	0.5	Complete	Colour and outermost point detection for both rucks and players are now combined.
Move repeating functions into the functions file	Liza	6-Apr	6-Apr	0.5	Complete	Moved colour detection functions into separate functions file to clean up the code
Research and write code for detecting contours from image	Liza	6-Apr	6-Apr	0.5	Complete	Written code to detect contours from an image, however field edges are not being detected as contours.
Work on the sprint document	Liza	6-Apr	6-Apr	1	Complete	Edited tasks
Readjust outermost detection to focus on legs / feet of the players	Liza	5-Apr	6-Apr	2	Complete	Adjusted the outermost point to only focus on the bottom half of the box. Currently outermost point is detecting arms as well. Will double check with Liam. As plan B, for now, created the bounding box detection of feet separately.
Further refine the ball model so that the model can detect the ball well in a further out view	Brooke	5-Apr	6-Apr	2	Complete	Further refine the ball model so that it can be used to detect the ball from far away. This is needed to be able to measure when the ball leaves the hands of the player
Create a burndown chart for Sprint 3	Liza	6-Apr	~	0.5	In Progress	Was waiting for other team members to indicate whether they have done the work. Moved to the next sprint. Won't affect project progression.
Upload signed feedback form to shared folder	Liam	28-Mar	~	0.5	In Progress	Awaiting Sam to email back the filled out form. Move to the next sprint. Moved to the next sprint. Won't affect project progression.
Draw a line for lineouts based on continuous detection of the hooker	Upite	6-Apr	~	3	In Progress	Won't be detecting off hooker, instead will go off lineout detection and use hooker to test accuracy of line drawing. Due to variance in angles, might have to train the dataset on model then import lines based on angle. Moved to next sprint. May affect project progress.

Draw line for lineout from point of detection	Upite	6-Apr	~	1	Incomplete	Still working on it, dependent on above detection. Moved to next sprint. May affect project progress.
Measure parameters of the field	Liam	6-Apr	~	3	Incomplete	Due to an assignment due, Liam was unable to begin work on this. Liza attempted to start this work by detecting contours from the image but was unable to measure parameters of the field. Moved to the next sprint, may affect project progress.
TOTAL POINTS	51.5 / 59.5 COMPLETED					
S04: Sprint 4 (Week 7 - Mid Semester Break)						
(from sprint 2) Measure and mark the time the ball leaves player's hands in lineout	Brooke	28-Mar	7-Apr	3	Complete	Started to work on the code for detecting when the ball leaves the hands. It was discovered that other models need to be further refined as the model is having a hard time detecting the ball from a faraway angle. Moved to the next sprint. May affect project progress.
Help Liza update sprint document on what has been completed	Brooke	7-Apr	7-Apr	0.5	Complete	So that all tasks were outlined within the sprint, I decided to go through and help Liza fill out the tasks which I completed in the sprint so that all details were there
(from sprint 3) Draw line for lineout from point of detection	Upite	6-Apr	7-Apr	1	Complete	Line was drawn from detection of lineout, diagonally across box to emulate offside midpoint line. However, because of direction and camera angle the line won't look as accurate. Have started hooker detection to try get a more accurate line
Get the project to be more accurate with marking where the ball has been released from hands	Brooke	7-Apr	7-Apr	1	Completed	The initial mark of when the ball was getting released was quite inaccurate. Further refinement of the code was implemented to get a more accurate time.
Further refinement of the ball model	Brooke	7-Apr	7-Apr	2	Completed	Further refinement of the ball model was needed so that it stops detecting objects that aren't the ball and only detects the ball. This most likely will still need more refinement in the future
Add hooker detection to lineout model	Upite	7-Apr	8-Apr	4	Complete	Added hooker as class to dataset then label/annotated the hooker and made/retrained models to improve confidence levels. Current model for lineout and hooker detection is v4; Model has been retrained.
Meeting with Dimitri	Brooke, Liam, Liza, Upite	11-Apr	11-Apr	1	Complete	Liam was unable to attend
Research Studio Huddle scripts and code	Upite	11-Apr	11-Apr	1	Complete	Research whether it will be possible to use Studio scripts and add our code to it. Found a lack of publicly available code and unclear on how to add to it.
Measure and draw lines 10m either side from lineout	Upite	6-Apr	~	0	Dependency Incomplete	Dependency task: complete in week 1. Moved into next week of the same sprint.
(from sprint 3) Draw a line for lineouts based on detection of the hooker	Upite	8-Apr	~	0	Dependency Incomplete	Dependency task: complete in week 1. Moved into next week of the same sprint.
Pose estimation for players feet	Liza	12-Apr	~	0	Dependency Incomplete	Dependency task: complete in week 1. As I was working on another assignment due this week, I didn't have enough time to put into finishing my tasks. Moved into next week of the same sprint.
Research how to improve person detection from far away angles	Liza	12-Apr	~	0	Dependency Incomplete	Dependency task: complete in week 1. As I was working on another assignment due this week, I didn't have enough time to put into finishing my tasks. Moved into next week of the same sprint.

Measure and draw lines between two objects	Upite	12-Apr	~	0	Dependency Incomplete	Dependency task: complete in week 1. https://www.youtube.com/watch?v=L8am1QoVn4. Moved into next week of the same sprint.
(from sprint 3) Measure parameters of the field	Liam	12-Apr	~	0	Dependency Incomplete	Dependency task: complete in week 1. Moved into next week of the same sprint.
Take notes and upload meeting minutes (Dimitri meeting)	Brooke	11-Apr	11-Apr	0	Reassigned	Reassigned to Liza as Brooke was using her computer to show our project on the big screen and was unable to type notes.
(from sprint 3) Attempting to draw field lines using the probabilistic Houghlines transform	Liam	14-Apr	14-Apr	2	Complete	Houghlines transform only detects straight lines. Program struggled with lines that weren't in the centre of the image
Creating functions to combine collinear lines so dotted 10m lines on the field can be combined into one line	Liam	15-Apr	15-Apr	2	Complete	Functions worked as intended to combine collinear lines, however, did not work on the 10m dotted lines.
Research image distortion to find out why collinear functions and line detection weren't worked as intended	Liam	16-Apr	16-Apr	0.5	Complete	Measuring the provided images, found that the lines towards the edge were not straight due to lens distortion
Research into OpenCV camera calibration to rectify lens distortion	Liam	16-Apr	16-Apr	1.5	Complete	Researched the OpenCV camera calibration options. All examples and functions utilized a chessboard to calibrate. Possible to calibrate using hardcoded points, but was unable to find how to dynamically find these points
Creating function to find intersection between two lines to account for image angles	Liam	16-Apr	16-Apr	1.5	Complete	To have offside lines straight across the field, needed to calculate angle relative to position in the image. For that, need the intersection point beyond the field of two lines on the field, then draw the a line from there to the detected offside point. The function takes two lines in (x1, y1, x2, y2) format and returns the intersection.
Get Liam to upload/link retrained model for lineup and hooker to repository	Upite	16-Apr	16-Apr	0.5	Complete	Model is retrained
Take notes and upload meeting minutes (Dimitri meeting)	Liza	11-Apr	17-Apr	0.5	Complete	Uploaded to shared document
(from sprint 3) Create a burndown chart for Sprint 3	Liza	17-Apr	17-Apr	0.5	Complete	Created burndown chart for sprint 3
Work on the sprint document	Liza	17-Apr	17-Apr	1	Complete	Work on Sprint 4 of the sprint document
(from week 1 of sprint 4) Pose estimation for players feet	Liza	17-Apr	17-Apr	0.5	Complete	Pose estimation now works for detecting specific body part in zoomed in view but not in wide angle. It uses a pre-trained data set from YOLO to do so.
Draw a bounding box around feet	Liza	17-Apr	17-Apr	0.5	Complete	Figured out a way to draw 2 bounding boxes, one for jersey detection and one for detecting feet. It is not very accurate though as it draw the box based off of the parameters of the whole person.
Attempt to manually train a dataset in Roboflow to detect feet	Liza	17-Apr	17-Apr	0.5	Complete	Attempted to train dataset detecting feet. After half an hour of work and only getting through 2 images, I realized this may not be feasible. Will have to find another way to detect feet from wide angle.

Make video display less busy	Liza	17-Apr	17-Apr	0.5	Complete	Even though players are still detected, the bounding box is not drawn (for the original code, though if we are using pose estimation, the bounding box is automatically drawn).
Change colour detection so it's not hard-coded but dynamic	Liza	17-Apr	17-Apr	1.5	Complete	Changed colour detection code so it compares all detected jerseys to all colours specified in a list. Then, assigns closest colour to each player. In this case, referees are detected.
Update colour sorting function to be more robust	Liam	17-Apr	17-Apr	0.5	Complete	Created new function to find 2 dominant colors and then compare to each detected colour to classify teams
Test different ways to make colour detection more accurate with wide angle view	Liza	17-Apr	17-Apr	0.5	Complete	Tried making ROI smaller and bigger, with a different combination of threshold, brightness and k values but accuracy seems to stay average to poor with wide angle views
Write code for perspective transform	Liam	17-Apr	17-Apr	0.5	Complete	Implemented code that can take points on a field and transform them to a 2D top-down view. Function works as intended but still requires the implementation of code to dynamically find relevant points on the field.
Test the difference between yolov8n and yolo11n pose estimation models	Liza	18-Apr	18-Apr	0.5	Complete	Noticed that yolov8n is just slightly better at detecting pose estimation from wide angle view, so decided to use this instead of the yolo11n version.
Created function to draw a straight line based on a detected contour	Liam	18-Apr	18-Apr	1	Complete	The function works as intended, however, still requires better input from the contour detection to get more accurate lines.
Extract coordinates for one player's left ankle from pose estimation, using an image	Liza	18-Apr	18-Apr	1	Complete	Able to print coordinates of the left and right ankles for one player in an image. Need to update for multiple players and then video.
Emailed Dimitri to set up a meeting	Liza	18-Apr	18-Apr	0.5	Complete	Emailed Dimitri to set up a meeting next week to ask some technical questions about line detection and offside lines
Extract coordinates for every player's left ankle from pose estimation, using an image	Liza	18-Apr	18-Apr	0.5	Complete	Able to print coordinates for all detected left ankles
Extract coordinates for every player's left and right ankles from pose estimation, using an image	Liza	19-Apr	19-Apr	0.5	Complete	Need to have both ankles so we can compare the x coordinates and see if they cross over / intercept the offside line from the ruck
Print leftmost and rightmost coordinates for the ruck	Liza	19-Apr	19-Apr	0.5	Complete	Able to extract the coordinates of where the circle is drawn on the left and rightmost edges of the ruck. Can use these to draw an offside line.
Modify pose estimation code to detect ankles from video and store results in a list	Liza	19-Apr	19-Apr	0.5	Complete	Detects ankles from a video and stores results in a list. Processing times are very slow though, so may need to find a way to process faster.
(from week 1 of sprint 4) Research how to improve person detection from far away angles	Liza	19-Apr	19-Apr	0.5	Complete	May need to fine tune the dataset to include examples of players from a wide angle. Apparently YOLOv9-E, may be better at handling smaller objects, so could test that.
Fix merge to get updated model and reconfigure lineout line code	Upite	19-Apr	19-Apr	0.5	Complete	Done
Tweak ball model to make it more accurate in detection	Brooke	18-Apr	19-Apr	3	Complete	Needed to work on the ball model as it was still having a hard time detecting it or it was still detecting other objects. Added 400 more images to the model dataset to help it detect the ball better. Need Liam to add new model to project

Perform and separate detection results for lineout and hooker then draw boxes and label each box for each if detected	Upite	20-Apr	20-Apr	1	Complete	Model detects hooker [0] and lineout [1] but detection (results) will have lineout [0] hooker [1]
Draw midline for offside across the field based off detection of centre of hooker	Upite	20-Apr	20-Apr	1	Complete	Used box centre from functions to detect centre then drew line to lineout corner. Will need to talk to Liam about using his intersection method to draw an accurate, parallel line
Review research and measure lines between two objects	Upite	20-Apr	20-Apr	1.5	Complete	Tried using Ultralytics distance calculation but had issues with the module. Unfortunately some example/methods requires knowing the real-life size of object. I've used Euclidean ('c' in Pythagoras theorem) to calculate the length of the line.
Convert measurement of units into yards	Upite	20-Apr	20-Apr	0.5	Complete	Looked up suggested conversion for pixels to yards. Converts length measured to yards. Will need to test for accuracy/check against reference point. https://www.deepconverters.com/typography/pixel-x-to-meter
Test accuracy of pixel to yard conversion	Upite	20-Apr	~	1	In Progress	COMMENTS. Moved into the next sprint. This won't impact the progress of our project.
Combine lineout offside line drawing and ball models together to detect offsidess	Upite	20-Apr	~	1	In Progress	Awaiting dependency- Copied lineout and ruck datasets and merged in Roboflow. Training new model off existing Lineout model and Ruck model, respectively. Will add ball detection dataset when it's ready. Moved into the next sprint. This won't impact the progress of our project.
(from sprint 3) Upload signed feedback form to shared folder sprint 3	Liam	15-Apr	~	0.5	In Progress	Awaiting Sam to respond. Moved into the next sprint. This won't impact the progress of our project.
Working on image line detection using contours	Liam	17-Apr	~	2	In Progress	Due to the limitations with the probabilistic Houghlines transform and image distortion, working on finding field lines using OpenCV contours. Currently able to find the field lines, still working on eliminating noise and working across more images. Moved into the next sprint. This won't impact the progress of our project.
Email Sam the feedback form + team progress for sprint 4	Liam	20-Apr	~	0.5	Incomplete	Dimitri and Jake are up to date with our progress, just need to email Sam as well and get him to sign the form. Moved into the next sprint. This won't impact the progress of our project.
Upload signed feedback form to shared folder for sprint 4	Liam	20-Apr	~	0.5	Incomplete	Need to email Sam to request form. Moved into the next sprint. This won't impact the progress of our project.
Ensure line drawn for lineout is correct angle	Upite	20-Apr	~	2	Incomplete	Awaiting dependency. Need to field detection and then can write algorithm for line angle or use calculate intersection point. Moved into the next sprint. This won't affect the progress of our project.
(from week 1 of sprint 4) Measure and draw lines 10m either side from lineout	Upite	20-Apr	~	1.5	Incomplete	COMMENTS. Moved into the next sprint. Even though we aimed to have this completed in this sprint, it shouldn't take too long to implement, and hence shouldn't affect project progress too much.
Reconfigure distance method to draw line after calculating distance	Upite	20-Apr	~	2	Incomplete	COMMENTS. Moved into the next sprint. This won't affect the progression of our project.
TOTAL POINTS	41.5 / 52.5 COMPLETED					
S05: Sprint 5 (Week 8 - 9)						
Update Roboflow model to include ref point as well	Brooke	21-Apr	21-Apr	3	Complete	It was recommended by Jake that we may be able to utilize the position of the ref to be able to determine the line of touch within the lineout

Send follow up email to Dimitri to schedule a meeting for this week	Liza	22-Apr	22-Apr	0.5	Complete	Email sent. Awaiting response.
Write and Upload meeting minutes with Wayne	Brooke	22-Apr	22-Apr	0.5	Complete	Uploaded meeting minutes to shared OneDrive folder
(from sprint 4) Email Sam the feedback form + team progress for sprint 4	Liam	20-Apr	22-Apr	0.5	Complete	Dimitri and Jake are up to date with our progress, just need to email Sam as well and get him to sign the form.
(from sprint 3) Upload signed feedback form to shared folder sprint 3	Liam	15-Apr	22-Apr	0.5	Complete	No response, confirmed with Wayne that not all feedback forms are needed
(from sprint 4) Upload signed feedback form to shared folder for sprint 4	Liam	20-Apr	22-Apr	0.5	Complete	Completed
Finalize artefact agreement	Liza, Liam	22-Apr	22-Apr	2	Complete	Finalized artefact agreement and sent to all shareholders
Attend meeting with Wayne	Brooke, Liam, Liza, Upite	22-Apr	22-Apr	0.5	Complete	All members present
Merge all work from Sprint 4 in GitHub	Liam	22-Apr	22-Apr	1	Complete	Merged completed work and cleaned up directory
Write function to allow file selection in file explorer instead of with hardcoded filenames	Liam	22-Apr	22-Apr	0.5	Complete	Instead of writing a file path or hardcoding the file path beforehand, used Tkinter to bring up standard file selection window
(from sprint 4) Combine lineout offside line drawing and ball models together to detect offsidess	Upite	20-Apr	22-Apr	2	Complete	Awaiting dependency- Copied lineout and ruck datasets and merged in Roboflow. Training new model off existing Lineout model and Ruck model, respectively. Will add ball detection dataset when it's ready. - This has now been completed
Change ball release code to reflect the new model and adjust annotations	Brooke	23-Apr	23-Apr	1	Complete	Due to the new model being trained where a couple of the smaller models were combined into the one large model, the code with in the ball release detection file needed to be adjusted to reflect this new instance
Work with the new model to display all detections of the ball, lineout, players and Ref.	Brooke	24-Apr	24-Apr	2	Complete	Changed the code to display all detections needed from the model.
Tweek the lineout file to get better accuracy of the ball release	Brooke	25-Apr	25-Apr	1	Complete	Updated
(from sprint 4) Refactor/reverse distance calculation code	Upite	20-Apr	26-Apr	0.5	Complete	Reconfigured function that converts distance (pixels to metres) to calculate amount of pixels in requested metres. This will be and was later used to request points/lines of various lengths and distances apart.
(from sprint 4) Test accuracy of pixel to yard conversion	Upite	23-Apr	26-Apr	1	Complete	Changed from yards to metres. Conversion was affected by resize of image and camera quality, angles and zoom. Not sure how to account for that.
Create sprint 4 burndown chart	Liza	27-Apr	27-Apr	0.5	Complete	Created in excel document

Combine pose estimation and ruck detection so results are displayed in the same window	Liza	27-Apr	27-Apr	0.5	Complete	Now instead of drawing a bounding box, 2 dots are drawn where the ankles of the players are located. A bounding box is still drawn around the ruck
To the task above, add the jersey colour detection code to ensure they all work together	Liza	27-Apr	27-Apr	1	Complete	Code now detects ROI from the jersey, and uses the dominant colour to draw circles around the player's ankles, instead of drawing the bounding box. Makes the view less cluttered.
Work on the sprint document	Liza	27-Apr	27-Apr	0.5	Complete	Update tasks
(from sprint 4) Working on image line detection using contours	Liam	17-Apr	28-Apr	6	Complete	The line detection now works as intended. Performance can vary depending on image lighting. Could look into adaptive thresholding again, however, this did not improve performance previously
Use new distance function to find (2) points either side of hooker centre (midline offside point)	Upite	26-Apr	28-Apr	1	Complete	Refactored distance calculator to draw lines of chosen distance. Code gets left and right points 10m from midline ensuring lines are perpendicular
Get midpoint of Ref position to use for the line	Brooke	28-Apr	28-Apr	1	Complete	It was determined the midpoint of the bottom of the bounding box for the ref to be able to be used for the line of touch within the lineout
Test field line detection with colour masks	Liza	29-Apr	29-Apr	0.5	Complete	Worked worse than current dilation image thresholding
Investigate Roboflow use for field line detection	Liza	29-Apr	29-Apr	0.5	Complete	Unable to find existing datasets to detect lines. May take too long to create own datasets.
Investigate Segment Anything Model (SAM) for field line detection	Liza	29-Apr	29-Apr	0.5	Complete	Unable to install and properly run on VSCode. Also required manual segmentation which will take the same amount of time as doing it in Roboflow. The segmentation is not as accurate.
Get the 3 points (midline, 10m left of midline, 10m right of midline) and draw 3 parallel lines	Upite	28-Apr	29-Apr	1	Complete	Still need to fix angle/point that lines draw to so that they visually appear parallel (mathematically I think they are parallel already).
(from week 1 of sprint 4) Measure and draw lines 10m either side from lineout	Upite	20-Apr	29-Apr	1.5	Complete	Lines are drawn to random point as intersect point/angle is dependent on field detection
Re-do code for detecting lineout ball and player detection to include functions	Brooke	2-May	2-May	1	Complete	Completed
Move functions from main file into functions file for lineout	Brooke	2-May	2-May	0.5	Complete	Transferred functions and changed main file for the lineout detection file
Refactored and combined previous code into functions	Brooke, Liam, Upite	2-May	2-May	3	Complete	Completed
Refactored and combined previous code into functions	Brooke, Liam	2-May	2-May	3	Complete	Completed

Refactored and combined previous code for lineout and beginning to look at detection	Liam, Upite	3-May	3-May	3	Complete	Combining field, intersect and offside (mid, left and right) code and can draw setup for lineout offside detection. However, distance is slightly off due to camera zoom. Started code for detecting whether players are offside in relation to a line
(from sprint 4) Ensure line drawn for lineout is correct angle	Upite	20-Apr	4-May	1	Complete	Using my previous code and Liam's intersection point and field code to get all lines (from 3 offside guidelines) correctly angled
Reformat Release Plan	Upite	3-May	~	1	In Progress	Release plan is written, just needs to be changed into suggested format. Moved to sprint 6. Doesn't impact progression of project.
Refactor code so that lineout offside lines (drawings) are applied to video instead of photo	Upite	3-May	~	1	In Progress	Talked to Liam and this will be better done later after everything is combined and working. Moved to sprint 6. Doesn't impact progression of project.
TOTAL POINTS	43 / 45 COMPLETED					
S06: Sprint 6 (Week 10 - 11)						
Email Dimitri about fixing distance	Upite	6-May	6-May	0.5	Complete	Sent.
Read through final report structure. Create Assessment 3 final report document, upload to shared folder and create chapters and headings for team and each chapter	Liza	6-May	6-May	1	Complete	Uploaded to Assessment description, rubric and guide folder
Create likelihood and impact metrics for risk registry	Liza	6-May	6-May	0.5	Complete	Completed and uploaded to appendix of final report document
Complete the team risk management section of final report	Liza	6-May	6-May	1	Complete	Completed under the Risk Management heading in final report document
Create burndown chart for Sprint 5	Liza	6-May	6-May	0.5	Complete	Completed in excel document under S05 Burndown tab
Refactoring ruck and player detection code into functions	Liam	6-May	6-May	1	Complete	Done
Updating ruck offside detection to incorporate ruck detection code	Liam	6-May	6-May	1	Complete	Code gets ruck points from previously implemented ruck detection results
Updating ruck offside detection to incorporate player detection code	Liam	6-May	6-May	2	Complete	Main implementation has been complete. Testing performance and working through errors
Attend meeting with Wayne	Brooke, Liam, Liza, Upite	6-May	6-May	0.5	Complete	All members present
Attend team meeting	Brooke, Liam, Liza, Upite	6-May	6-May	1	Complete	All members present

Write the meeting minutes from meeting with Wayne	Brooke	7-May	7-May	0.5	Complete	Meeting minutes were recorded and uploaded to shared folder
Look at offside application for all types	Upite	5-May	7-May	1	Complete	Done
Continue to refine the models by adding more images. Add for Wide angle	Brooke	7-May	8-May	2	Complete	Complete
(from sprint 5) Refactor code so that lineout offside lines (drawings) are applied to video instead of photo	Upite	3-May	8-May	1	Complete	Awaiting completion of combining code to ensure they work together before beginning this.
(from sprint 5) Reformat Release Plan	Upite	3-May	8-May	1	Complete	Done, in Excel
Refactoring lineout code with hooker code	Upite	6-May	9-May	1.5	Complete	Refactoring lineout code to use model detection instead of hardcoded test coordinates. May need to retrain model before combining all models
Finalize (finish writing and editing) risk management section of A3 - final report.	Liza	13-May	13-May	1	Complete	Available in final report document
Write 'Project Context, Goals, Scope' team section for introduction in final report	Liza	14-May	14-May	1	Complete	Available in final report document
Write 'Project Outcomes and Successes' team section for introduction in final report	Liza	14-May	14-May	1.5	Complete	Available in final report document
Update and add to the communication plan table	Liza	14-May	14-May	0.5	Complete	Available in final report document appendix
Write 'Communication Plan' team section in final report	Liza	14-May	14-May	1	Complete	Available in final report document
Meeting with Dimitri	Upite, Liam	15-May	15-May	2	Complete	Discussed how to fix the slight inaccuracy of the meter measurement
Email Sam the feedback form	Liam	16-May	16-May	0.5	Complete	Done
Minutes for meeting with Dimitri	Upite	16-May	16-May	0.5	Complete	Done
TOTAL POINTS	24 / 24 COMPLETED					

S07: Sprint 7 (Week 12 - 13)						
Reorganize user stories	Upite	20-May	20-May	1.5	Complete	Assigning user stories/task ID under epics, to match Release Plan
Attend meeting with Dimitri	Liam, Upite	20-May	20-May	1	Complete	Get guidance on line and field line detection
Improve distance calculation by sampling image quality (ppi/dpi)	Upite	20-May	20-May	1.5	Complete	Did not work
Record start and end scenes for final video	Brooke, Liam, Liza, Upite	23-May	23-May	2	Complete	Done
Write 'Client Relationship and Expectations' team section in final report	Liza	26-May	26-May	1.5	Complete	In shared folder
Write 'Team Collaboration' team section in final report	Liza	26-May	26-May	1.5	Complete	In shared folder
Write 'Project Management Approach' team section in final report	Liza	27-May	27-May	1.5	Complete	In shared folder
Clean up and add comments to all code	Liza	30-May	30-May	2	Complete	Done
Add unit testing for all files and functions	Liza	30-May	30-May	2	Complete	Done
Implementing manual mode	Liam	28-May	30-May	3	Complete	Done
Combine all completed code	Liam	16-May	30-May	30	Complete	Done
Write Roboflow instructions	Brooke, Liam	30-May	30-May	1.5	Complete	Completed Brooke's part, waiting for Liam to include how to train the model. Completed and uploaded
In person team meeting to work on assignment	Liam, Liza	30-May	30-May	12	Complete	Met at university to work on code and final report
Edit the appendices and add project reference appendix with paths to files	Liza	31-May	31-May	2	Complete	Done

Modify Roboflow instructions and add more clarify to each step	Liza	31-May	31-May	1	Complete	Done
Edit and proofread all project management documents	Liza	31-May	31-May	3	Complete	Done
Create submission folders and organize according to submission guide	Liza	31-May	31-May	0.5	Complete	In OneDrive
Create burndown chart for sprint 6	Liza	31-May	31-May	0.5	Complete	In Excel
Additional model training and implementation	Liam	23-May	31-May	10	Complete	Done
Create burndown chart for sprint 7	Liza	31-May	31-May	0.5	Complete	In Excel
Write troubleshooting documentation	Upite	29-Jul	31-May	1.5	Complete	In OneDrive
Write installation instructions	Upite	29-May	31-May	1.5	Complete	In OneDrive
In person meeting to work on assignment	Liam, Liza	31-May	31-May	12	Complete	Met at university to work on code, final report and finalizing project management documents
Finalize all code prior to filming video	Liam	31-May	1-Jun	15	Complete	Done
Film screen recording part of the final video	Liam	1-Jun	1-Jun	1	Complete	Done
Edit and put together final video	Brooke	1-Jun	1-Jun	1	Complete	Done
Work on individual chapters of the report	Brooke, Liam, Liza, Upite	29-May	1-Jun	8	Complete	Done
Write testing documentation	Brooke	28-May	1-Jun	2	Complete	In OneDrive
Readjust and reformat user stories, initial velocity and release plans	Upite	1-Jun	1-Jun	2	Complete	In Excel
TOTAL POINTS	122.5 / 122.5 COMPLETED					